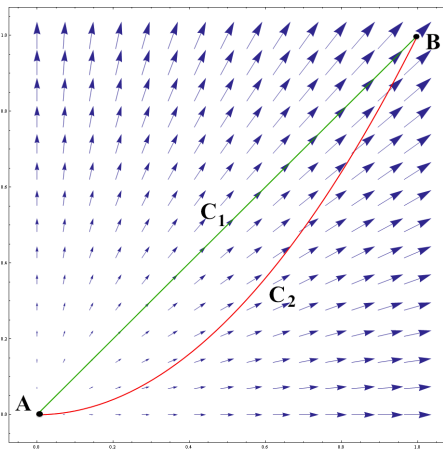
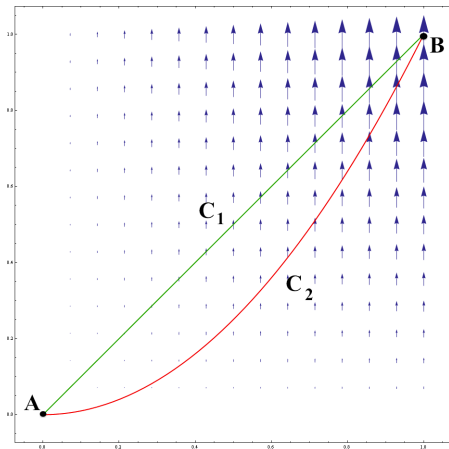


Topic: Path Independence

I. Introduction

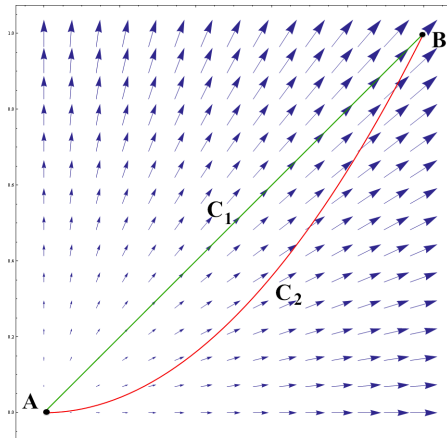
Question: Does the amount of work required to move a mass from A to B in a force field depend on the path you take?



Answer:

II. Path Independence

Defn: Let C_1 be a curve from point A to B . The line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ is said to be **path independent** if



III. Conservative Vector Fields and Potential Functions

Defn: A vector field $\mathbf{F}$ is **conservative** if

Thm: If $\mathbf{F}$ is a conservative vector field then

Defn: The f above is called a

Ex: Is $f(x, y, z) = x \sin(z^2) + e^{-y} + 13y$ a potential for $\mathbf{F}(x, y, z) = [\sin(z^2), -e^{-y} + 13, 2xz \cos(z^2)]$?
Is $\mathbf{F}(x, y, z)$ conservative?

Ex: Is $f(x, y, z) = xy \ln(y) + z$ a potential for $\mathbf{F}(x, y, z) = \left[y \ln(y), \frac{x}{y}, 1 \right]$? Is $\mathbf{F}(x, y, z)$ conservative?

Thm: Let C be a curve from A to B . If $\mathbf{F}$ is conservative (i.e. $\mathbf{F} = \nabla f$ for some f), then

Idea of Proof:

There is a nice consequence of the previous theorem that is important in future discussions.

First a definition:

Defn: A curve, C , is **closed** if it has $\mathbf{a}$ and $\mathbf{b}$. Any point on the curve can be thought of as an $\mathbf{a}$ and an $\mathbf{b}$.

Thm: If $\mathbf{F}$ is **conservative** ($\mathbf{F} = \nabla f$) and C is closed then

Idea of Proof:

Then from the previous theorem, we have

Ex: The vector field $\mathbf{F} = [yz, xz, xy]$ is conservative and has potential $f(x, y, z) = xyz$ (i.e. $\mathbf{F} = \nabla f$). Find the work done by $\mathbf{F}$ on a mass along a curve, C , between $A(-1, 3, 9)$ and $B(1, 6, -4)$.

Soln:

IV. Finding Potential Functions

If we are given a conservative vector field $\mathbf{F}$, then we know that there must be a corresponding potential function f such that $\nabla f = \mathbf{F}$.

But what is f ?

Let $\mathbf{F} = [M, N, P]$ (Note: M, N, P are all functions of)

Then

So to find f , we must compute a sequence of

(i)

(ii)

(iii)

Ex: Let $\mathbf{F} = [e^x \cos y + yz, xz - e^x \sin y, xy + z]$

(a) Show $\mathbf{F}$ is conservative by finding an f such that $\mathbf{F} = \nabla f$

(b) Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is a curve from $A : (0, 0, 2)$ to $B : (2, \pi/2, 0)$

Ex: Let $\mathbf{F} = [y, x, 4]$.

(a) Show $\mathbf{F}$ is conservative by finding an f such that $\mathbf{F} = \nabla f$

(b) Compute $\int_{(1,1,1)}^{(2,3,-1)} y \, dx + x \, dy + 4 \, dz$

From part (a), we found that $f(x, y, z) =$

and therefore

V. Conservative Test for Vector Fields

In general, to show a vector field is conservative by finding f such that $\mathbf{F} = \nabla f$ can be time consuming and challenging.

Fortunately, there is a test to determine whether a vector field, $\mathbf{F}$, is conservative.

Conservative vector field test: Let $\mathbf{F} = [M, N, P]$ be a vector field.

Assuming the partial derivatives of M, N, P exist, then $\mathbf{F}$ is conservative if

This is also equivalent to showing

Ex: Is $\mathbf{F} = [2x - 3, -z, \cos z]$ conservative?

Since

Ex: Let $\mathbf{F} = [2xz, 3z, x^2 + 3y]$. Is $\mathbf{F}$ is conservative?

Ex: Let $f = x^2z + 3yz$ and $\mathbf{F} = 2xz\mathbf{i} + 3z\mathbf{j} + (x^2 + 3y)\mathbf{k}$.

(a) Is f the potential for $\mathbf{F}$?

(b) Suppose you know the work required to move a mass through $\mathbf{F}$ along a curve between the points $(1, 1, 1)$ and $(d, 2d, 2)$ is $10 \text{ N} \cdot \text{m}$.

What are the possible terminal points, $(d, 2d, 2)$, for C ?